

Kumbha Nagaswetha

Senior Data Scientist at Atlas Copco Group
M.Tech from Indian Institute of Science (IISc), Bangalore
Specialization in Deep Learning, Computer Vision

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Education

- **Indian Institute of Science, Bengaluru** 2020-2022
M.Tech in Signal Processing (Specialization in Deep Learning and Computer Vision) - CGPA : 7.0 Bengaluru, KA
- **Jawaharlal Nehru Technological University Kakinada** 2016-2020
B.Tech in Electronics and Communication Engineering - CGPA : 8.27 Kakinada, AP

Experience

- **Atlas Copco (India), Pune** 2022 - Present
Senior Data Scientist with 4 years of experience, specializing in Deep Learning, Computer Vision, and Optimization Techniques. Skilled in developing end-to-end ML pipelines for real-world applications, including machine health monitoring, survival analysis, surrogate modeling of physical processes. Proficient across deep learning architectures (CNNs, GANs, Transformers, Vision Transformers, Graph Neural Networks) and scientific machine learning methods (Neural Operators, Physics-Informed Neural Networks), with strong foundations in Signal Processing, Image Processing, Optimization, Anomaly Detection, and Domain Adaptation, with expertise in leveraging techniques like t-SNE and SHAP for model interpretability.

Projects - Deep Learning

- **Domain Adaptation in Machine Health Monitoring** Atlas Copco, June.2024–Present
 - Developed a domain adaptation framework for compressor valve fault diagnosis under varying operating conditions using vibration signals and limited target-domain data.
 - Investigated GANs, autoencoders, VAEs, and feature alignment techniques, achieving approximately 96% fault detection accuracy.
 - Employed t-SNE and SHAP to interpret CNN-based model decisions and gain insights into learned features.
- **AI based Design Optimization** Atlas Copco, November.2024–August.2025
 - Implemented the Bayesian method with multi-surrogate method for shape optimization of Jet Pump to find the best shape with least number of function evaluation and parallel computation in HPC cluster.
 - Utilized the Constant Liar approach with Kriging, RBNN, Random Forest, and SVR surrogates, achieving the best efficiency with minimal iterations. Best shape found within 30-40 iterations.
 - Built a custom library to perform Ordinary Kriging in n-dimensions, extending capabilities beyond existing implementations and enabling flexible surrogate modeling for complex design spaces.
 - Explored, compared and documented pros and cons of sampling methods (LHS, CVT, McQueen, etc.), surrogate models (Kriging, Radial Basis Functions, NNs, etc.) and optimization techniques (Bayesian, GA, PSO, BFGS) for optimization problems and used the PRESS (Predicted Residual Error Sum of Squares) method to select the best surrogate model for a given dataset.
- **Efficient Conditioning: A Theory of Pseudo-Observation Batch Bayesian Optimization** 2024–2025
 - Identified efficient conditioning as the structural property explaining why Constant Liar, Kriging Believer, and fantasy models generate diverse batches with Gaussian Process surrogates while parametric surrogates collapse.
 - Derived theoretical guarantees including an $\Omega(\ell)$ variance suppression radius and unified pseudo-observation methods under a common conditioning framework.
 - Introduced the Structural Diversity Diagnostic (SDD) to isolate surrogate compatibility from optimizer randomness; validated theory on Hartmann-6D, Ackley-8D, Levy-10D, and SVM hyperparameter tuning: CL/KB matches or outperforms explicit LP and is competitive with joint q-EI, qlogEI within noise.
 - Manuscript submitted to NeurIPS 2026.
- **Speed Estimation of a Compressor from Vibration Data** Atlas Copco, Sep.2022–March.2023
 - Developed a vibration-based compressor speed estimation pipeline using 1D CNNs, LightGBM, and feature engineering.
 - Achieved 99.5% classification accuracy after evaluating PCA, Random Forest, SVM, and neural network baselines.
- **Time Series Forecasting and Demand Planning** Atlas Copco, August.2023–November.2023
 - Developed demand forecasting models using ARIMA, SARIMAX, LSTM, N-HiTS, and Temporal Convolutional Networks.
 - Achieved 90% forecasting accuracy and 87% opportunity-to-sales prediction accuracy using external economic indicators.
- **Survival Analysis for Predictive Maintenance** Atlas Copco, April.2023– July.2023
 - Built survival analysis models for predictive maintenance using Random Survival Forests in R.
 - Developed an interactive Shiny dashboard for real-time prediction and visualization.

Projects - Computer Vision

- **Lubricant Level Estimation Using CNN on Audio Signals** Atlas Copco, March.2023–Sep.2023
 - Led the development and implementation of the CNN architecture for Mel spectrogram analysis.
 - Explored Vision Transformers, Synthetic Minority Oversampling Technique (SMOTE), data augmentation methods such as solarization, time masking, frequency masking, Gaussian blur, and experimented with weighted convolutional networks and adaptive learning rate strategies to improve robustness.
 - Utilized Python, TensorFlow, and Keras for model training, with Librosa for audio processing and VGG16 for feature extraction; achieved an R^2 score of 0.92 on the test set and deployed a pyfunc-based pipeline with a user-friendly interface.
- **U-FNO-DA: Fourier Neural Operators for Robust Domain Adaptation** 2025–2026
 - Designed U-FNO-DA, an architecture coupling Fourier Neural Operator (FNO) blocks for spectral learning, a UNet path for multi-scale spatial context, and CORAL feature alignment for cross-condition transfer; motivated by the observation that CNN filters are implicitly calibrated to the training-time frequency resolution and harmonic amplitudes, whereas bearing fault harmonics are globally structured in frequency space.
 - Validated on the Paderborn University bearing dataset under two deployment shifts, sensor-quality (train 64 kHz, test at 32/16 kHz) and operating-condition (limited target labels, $f = 0.1\text{--}0.9$) with a calibrated mel-spectrogram protocol that preserves harmonic positions across sampling rates.
 - U-FNO-DA achieved 0 training collapses across 51 runs versus 6/51 for a controlled CNN-DA baseline under identical hyperparameters, and learned reliably with as few as 10% of target-condition labels (0.74 mean target accuracy, 3 seeds); ablation confirmed both spectral and multi-scale spatial paths are necessary.
- **HDR Image Reconstruction using Un-supervised Learning - arXiv Publication** IISc, Dec 2021 - June 2022
 - Developed an unsupervised framework for HDR reconstruction from multi-exposure images without HDR supervision.
 - Proposed a novel structural-perceptual loss, improving visual quality and MEF-SSIM over baseline methods.
- **Facial Emotion Recognition** IISc, Feb 2021 - Jun 2021
 - Built facial emotion recognition models using AlexNet feature extraction and SVM classification on JAFFE, AFED, and FER datasets
 - Studied transfer learning and cross-dataset generalization.
- **A Fuzzy Based DWM filter For Impulse Noise Removal** JNTUK, Dec 2019 - Mar 2020
 - Designed a fuzzy directional weighted median filter for impulse noise removal while preserving edges.
 - Achieved improved MAE and PSNR compared with conventional median filtering methods.
- **2D Transformations for Signal Representation** Atlas Copco, April.2023
 - Compared spectrograms, Mel spectrograms, SSQ-STFT, wavelets, and Hilbert–Huang transforms for vibration and audio signal representation.
 - Evaluated their suitability for deep learning-based fault diagnosis.
- **Other Projects in IISc**
 - Implemented SIFT, image classification, semantic segmentation, BM3D denoising, JPEG compression, and perceptual image quality assessment.
 - Evaluated classical and deep learning approaches for image restoration, segmentation, and recognition tasks.

Technical Skills

Technology: Machine Learning, Deep Learning, Computer Vision, Image Processing, Signal Processing, HPC, Databricks

Languages: Python, C, MATLAB, PySpark

Libraries: OpenCV, Tensorflow, Pytorch, scikit-learn, scipy, imblearn, Pyspark, Pandas, NumPy etc.

Major Courses Undertaken: Digital Image processing, Advanced Image Processing, Computer Vision, Pattern Recognition and Neural Networks.

Extra-Curricular Activities

- Participated in internship at BSNL Telecommunications in Kakinada.
- Co-ordinator of the technical event related to the department of ECE in the Annual techno-cultural fest, JNTU Kakinada, 2020.
- Attained yellow grade in Taekwondo federation of India.
- Participated in department level chess competitions.